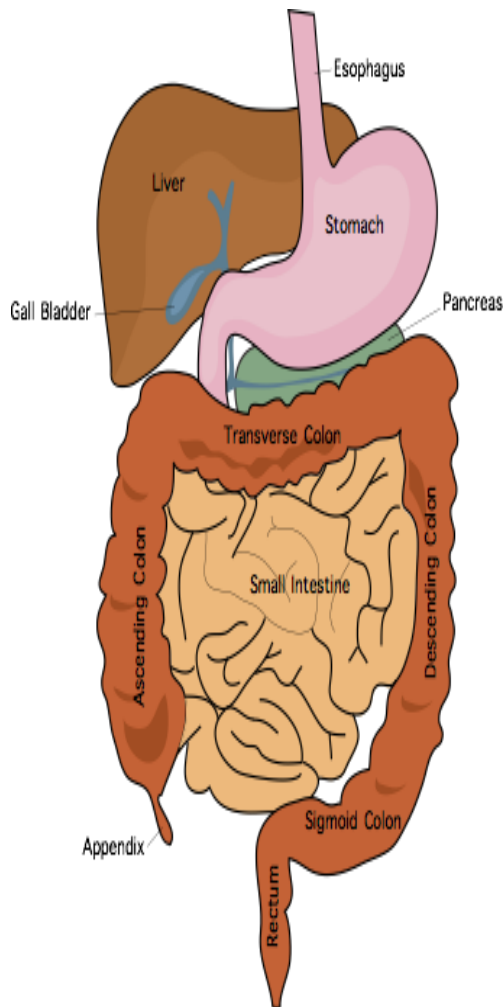


KWA-THEMA SKILL SCHOOL

NS-TECH WORK BOOK

YEAR 3 TERM 1

THEME: LIFE AND LIVING



WATER



MINERALS



VITAMINS



CARBOHYDRATES



PROTEIN



FAT



**FROM A FAMILY BLOG:
SETTLEINELPASO.COM**

NAME: _____

YEAR: _____

DIGESTIVE SYSTEM

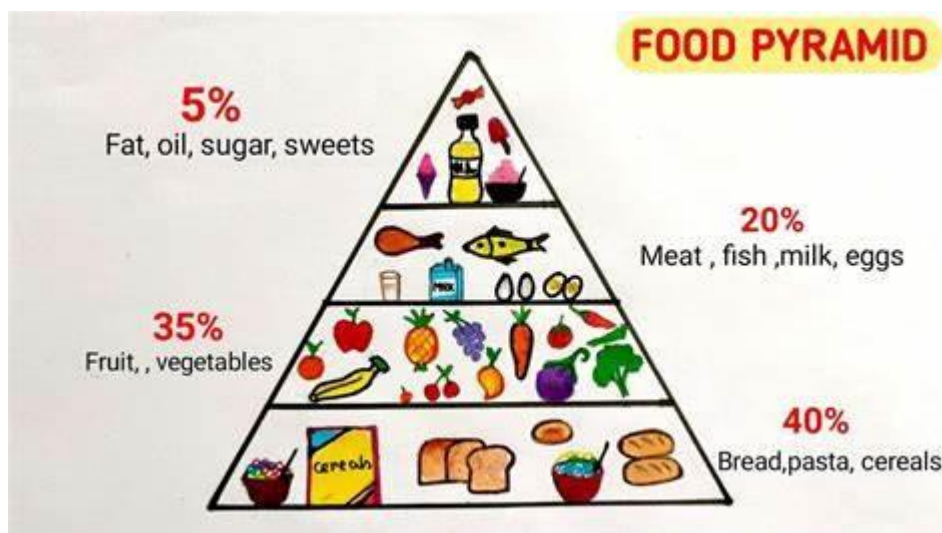
EXPLAIN ALL THE CONCEPTS (FOOD PYRAMID, FOOD GROUP, SUPPLIMENTS, FAT, NUTRITION, WATER, VITAMINS, MINERALS, CARBOHYDRATES, PROTEIN

FOOD PYRAMID

- The Food Guide Pyramid is an outline of what to eat each day based on the Dietary Guidelines. It provides a general guide that lets you choose a healthy diet that's right for you. The Pyramid calls for eating a variety of foods to get the nutrients you need and at the same time, the right amount of calories to maintaining healthy and physical wellbeing.

02/10/14

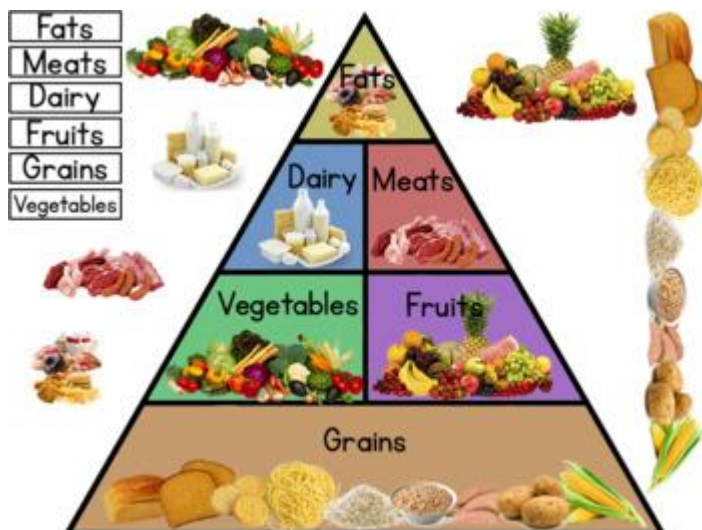
46



The 5 Basic Food Groups



HowforKids



Date _____ Name _____

FOOD GROUPS

Write down three examples of foods from the food groups below. In the last box, write your three favourite foods.

FRUITS & VEGETABLES

CARBOHYDRATES

PROTEIN

DAIRY

FATS

MY FAVOURITE FOODS

FRUITS &		VERGETABLE	CARBOHYDRATES		
PROTEIN			DAIRY		
FATS			MY FAVOURITES		

(10)

WHAT IS A DIETARY SUPPLEMENT

- ✘ Congress defined the term "dietary supplement" in the Dietary Supplement Health and Education Act (DSHEA) of 1994.
- ✘ A dietary supplement, also known as food supplement or nutritional supplement.
- ✘ Such as vitamins, minerals, fibers, fatty acids or amino acid, that may be missing or may not be consumed in sufficient quantities in a person's diet.



FATS



Healthy Fats



Healthy Diet

What is your idea of a healthy diet?

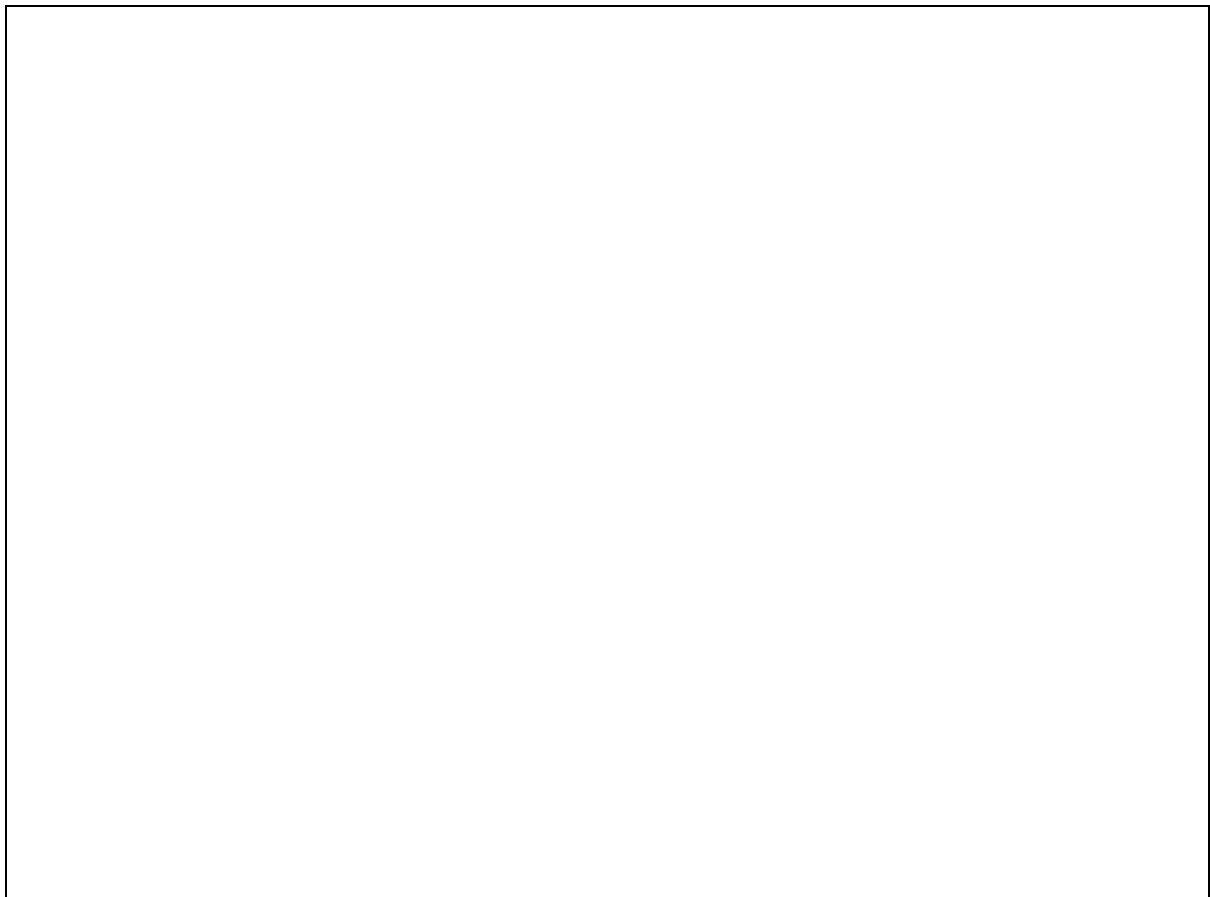
Complete the table below showing all the nutrients that the body needs.

Nutrient	Good For	Found in which Foods?
Carbohydrate s		
Lipids (Fats or Oils)		
Proteins		
Vitamins		
Minerals		
Fibre		
Water		

nutrients	good	Which food

(14)

FOOD PYRAMID TO BE DRAWN AND FILLED IN THE SPACE



**NUTRITION
DEFINATION AND ITS
COMPONENTS**

White Bread

Nutrition Facts	
Serving Size 1 slice (34g)	
Servings Per Container 18	
Amount Per Serving	
Calories 90	Calories from Fat 15
%Daily Value*	
Total Fat 1.5g	2%
Saturated Fat 0g	0%
Trans Fat 0g	0%
Cholesterol 0mg	0%
Sodium 140 mg	6%
Total Carbohydrate 16 g	5%
Dietary Fiber 1g	0%
Sugars 2g	
Protein 2g	
Vitamin A 0%	Vitamin C 0%
Calcium 4%	Iron 6%
* Percent Daily Values are based on a 2,000 calorie diet.	

Ingredients:
Enriched wheat flour (wheat flour, barley malt, niacin, iron, thiamine, mononitrate, riboflavin, folic acid), whey, high fructose corn syrup...

Whole-wheat Bread

Nutrition Facts	
Serving Size 1 slice (38g)	
Servings Per Container 18	
Amount Per Serving	
Calories 90	Calories from Fat 10
%Daily Value*	
Total Fat 1g	2%
Saturated Fat 0g	0%
Trans Fat 0g	0%
Cholesterol 0mg	0%
Sodium 190 mg	8%
Total Carbohydrate 18 g	6%
Dietary Fiber 3g	12%
Sugars 3g	
Protein 4g	
Vitamin A 0%	Vitamin C 0%
Calcium 4%	Iron 6%
* Percent Daily Values are based on a 2,000 calorie diet.	

Ingredients:
Whole-wheat flour, water, brown sugar, wheat gluten, soybean oil, wheat bran, yeast, salt, wheat germ, molasses, honey...



Definition of Nutrition

- Nutrition is the science of foods
- Involves the action, interaction, and balance of nutrients in relationship to health and disease
- Involves the processes by which food is digested, absorbed, transported, utilised, and excreted in the body
- Nutrition is concerned with social, economic, cultural, and psychological implications of food and eating

Nutrition

Nutrition: the science that studies food and how food nourishes our bodies and influences our health

Undernutrition: a diet that lacks energy or specific essential nutrients

Overnutrition: a diet that has an imbalance of fats, carbohydrates and proteins or simply too much energy

Malnutrition: any condition associated with undernutrition or overnutrition



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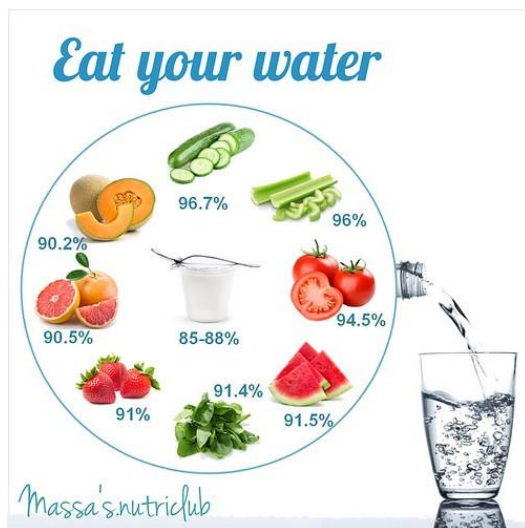
What are the reason we have to check nutritional value/contents on some of the food we buy (10)

[illegible]

Water

- All foods contain at least some water
- *Free water*
 - Held inside cells
 - Maintains properties of free water
 - May be removed by pressure
- *Bound water*
 - Is part of molecule structure
 - Reduced mobility
 - Does not retain properties of free water

NAME FOOD WTH WATER PERCENTAGE



ADD THE LIST :EXPANDED

Uitamins and Minerals

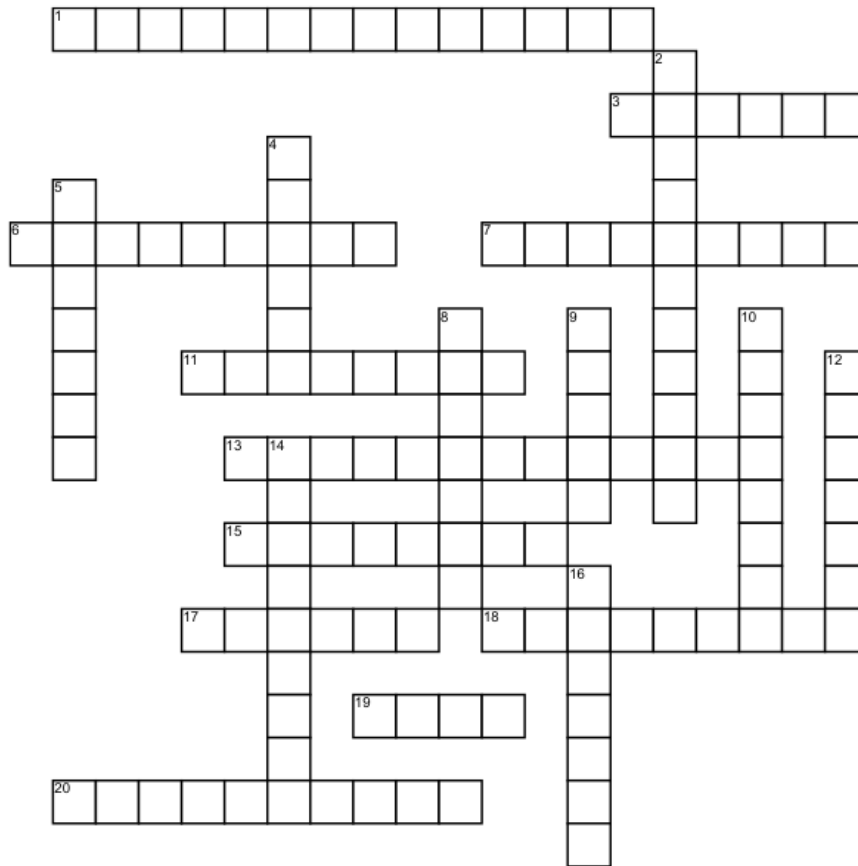


**ON THIS GROUP(FRUITS AND VERGETABLE) WE GET
VITAMIN AND MINERALS**

FOOD GROUP PUZZLE

Name: _____ Date: _____

Food Crossword Puzzle



Across

- 1. a place where you buy groceries
- 3. what water does when it is heated
- 6. a red fruit
- 7. the ketchup
- 11. something you can add to your drink
- 13. components used to make a food
- 15. the cake

- 17. a common drink at Starbucks

- 18. tasty; appetizing
- 19. a drink that can be sweet or iced
- 20. breakfast

Down

- 2. someone who only eats vegetables
- 4. the water
- 5. the strawberry
- 8. the fruit

- 9. a bitter taste in some candy

- 10. the ice cream
- 12. the cheese
- 14. a green, leafy vegetable
- 16. used to make lemonade

1ST TRY AND FIND ANSWERS

--	--	--	--

FUNCTIONS OF VARIOUS FOOD GROUPS

Carbohydrates	<i>Bread, pasta, potato</i>	Provides energy for our body.
Proteins	<i>Meat, fish, eggs, beans</i>	Needed for growth and repair.
Fats	<i>Butter, oil, nuts</i>	Provides energy, stores energy and insulates our body
Vitamins	<i>Fruit, vegetables</i>	Needed in small amounts to maintain health.
Minerals	<i>Red meat, milk, salt</i>	Needed in small amounts to maintain health.
Fibre	<i>Cereals, brown bread/rice, vegetables</i>	Helps our digestive tract work properly.
Water	<i>Drinks, in foods</i>	Needed for cells and body fluid.

HOW NUTRITION CAN LEAD TO DESEASE RESEARCH

USE THE SPACE BELOW(5)

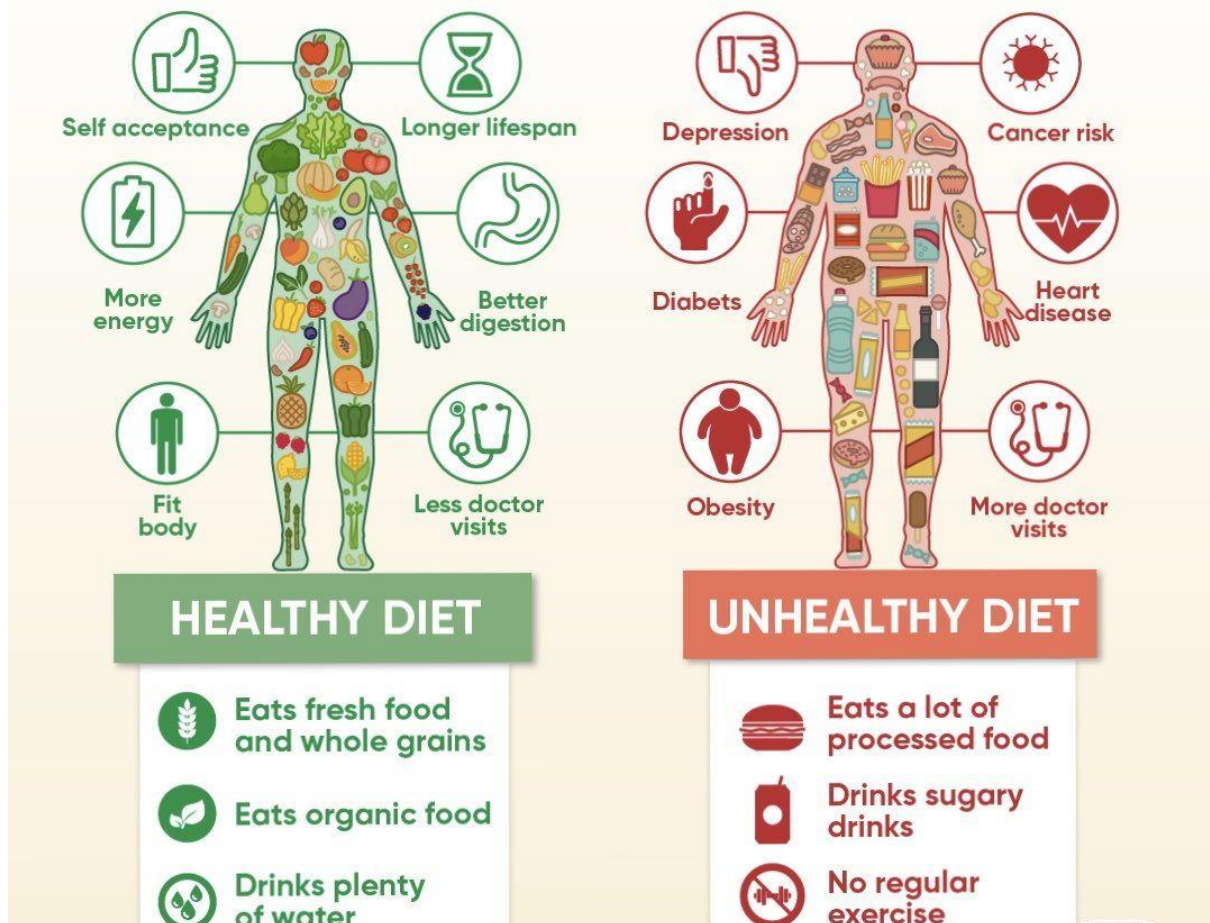
POOR NUTRITION

How Does Poor Nutrition Affect Us?



POOR NUTRITION VS GOOD NUTRITION: OWN INVESTIGATION (5)

HEALTHY DIET VS UNHEALTHY DIET



LOOK AT THE ABOVE PICTURE AND NAME ALL(10)

HEALTHY DIET FOOD LIST	UNHEALTHY LIST
1.	1.
2.	2.
3.	3.
4.	4.
5.	5.

FOOD PROCESSING

EXPLAIN

**(PROCESSING, PRESE
RVE, ARTIFICIAL, PRE
SERVATIVES, SUPPLI
MENTS**

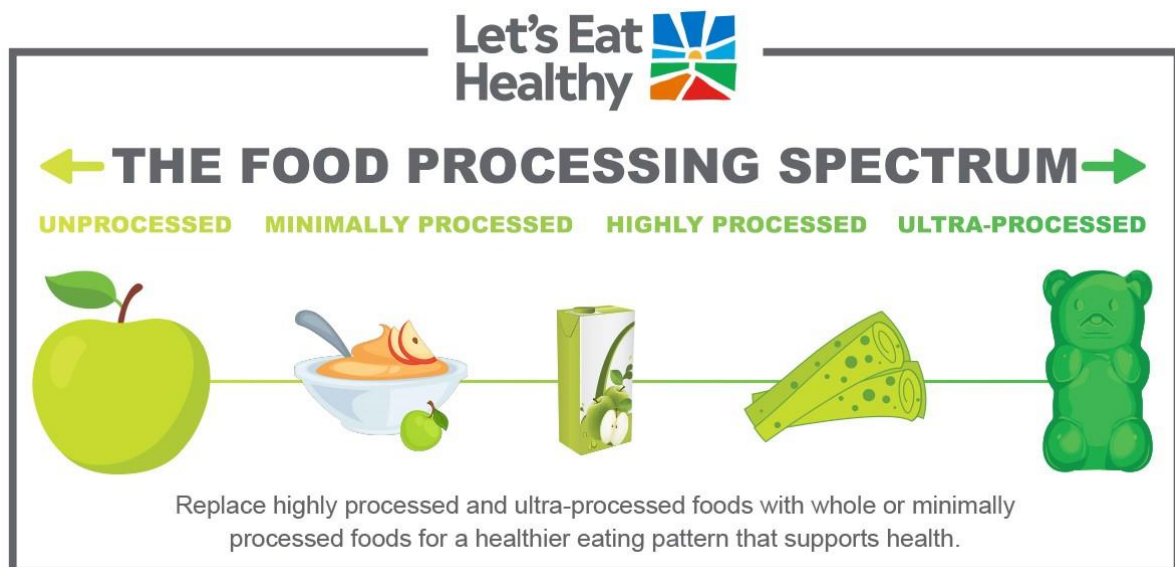
PROCESSING

Definition

- A common and traditional definition of Food Processing industry refers to that part of manufacturing that processes raw materials and intermediate products derived from the agricultural sector.
- Food Processing industry thus means transforming products originating from agriculture, forestry and fisheries.

Examples

- The industries that use agricultural, fishery and forest products as raw materials comprise a very varied group. They range from simple preservation (such as sun drying) and operations closely related to harvesting to the production, by modern, capital-intensive methods, of such articles as textiles, pulp and paper.



Businesses use food processing techniques to transform raw foods and ingredients into new products. These techniques run the gamut and include **milling grains into flour, crushing seeds to extract their oil, churning milk into butter, mixing ingredients to make batter, and baking cookies on an assembly line.**

Food Processing | Food System Primer



How food is processed in a human body

PARTS OF THE DIGESTIVE SYSTEM

1.

2.

3.

4.

5.

6.

☆ Label the parts of the digestive system in the diagram above and colour in using the colours given in the table on the right

WHAT DO THE PARTS OF THE DIGESTIVE SYSTEM DO?

☆ Complete the table below with the function of the main parts of the digestive system

Part	Function in Digestion
Mouth (Colour pink)	
Tongue (Colour red)	
Oesophagus (Colour orange)	
Stomach (Colour green)	
Small Intestine (Colour blue)	
Large Intestine (Colour purple)	

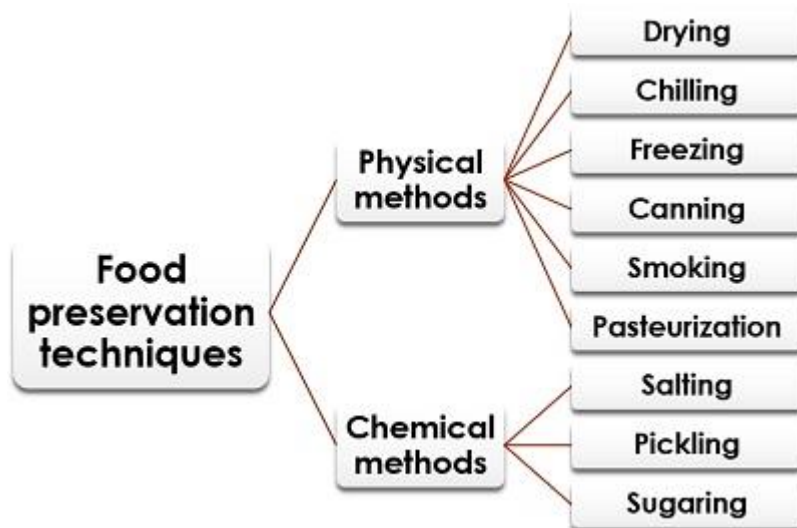
PARTS	FUNCTION IN DIGESTIVE

(10)

LOOK AT THE ABOVE PICTURE FROM THE FACTORY AND EXPLAIN HOW VORSE ROLLS ARE PREPARED (5)

INGREDIENTS/EQUIPMENTS	PROCESSED

HOW WE PRESERVE FOOD



SMOKING

- is a complementary process to curing that improves flavor and appearance, and can also act as a drying agent.
- Smoked meats are less likely to turn rancid or grow mold than unsmoked meats.



FREEZING

- is the process of chilling foods to at least 0°F.
- It can be used with all foods
- True freezing is not possible in the freezer compartment of your refrigerator where the temperature is typically much warmer, between 10°F to 32°F.
- Easy method



FOOD PRESERVATION METHODS

①



②



③



④

① Drying | ② Pickling | ③ Sugar / Salting | ④ Cooling

WHY FOOD IS PRESERVED

Food preservation is the process of preventing the decay and spoilage of produced fresh foods and storing it under prescribed conditions for future use.

Preserving foods plays an essential role in people's businesses and consumers. Marketers are in a position to earn more since it reduces food spoilage and decay.

It also stabilises food prices since prices tend to rise when supply is low and demand high

Name _____ Date _____

Preserving Food

What does preserving food mean ?

There are various methods of preserving food. From the helper box, select the type of preserving which best fits each description.

freezing drying canning pickling vacuum packing

Food is processed & sealed in an airtight container. _____

Food is dried. _____

Food is rapidly frozen. _____

Soaking food in salt water , vinegar or oil _____

Food stored in a airtight bag under vacuum _____

Identify the preserving methods



ANSWERS

1.

2.

3.

(12

EXPANDED

Label the preservation method.

Drag and match the correct description and example to match the method.

Preservation Method	Description	Example
	Drying food with smoke for a long period of time.	Meat Fish Banana
	Coating food with liquid wax.	Fish Chicken curry Ketchup Soy sauce
	Rubbing food with a lot of salt.	Apple Grape Orange Lemon
	Heating food to a certain temperature for some time followed by rapid cooling.	Milk Juice Milk products Cream
	Heating food at boiling temperature then put them in airtight container.	Fish Egg Vegetables

WHY DO WE ADD ARTIFICIALS IN FOOD?

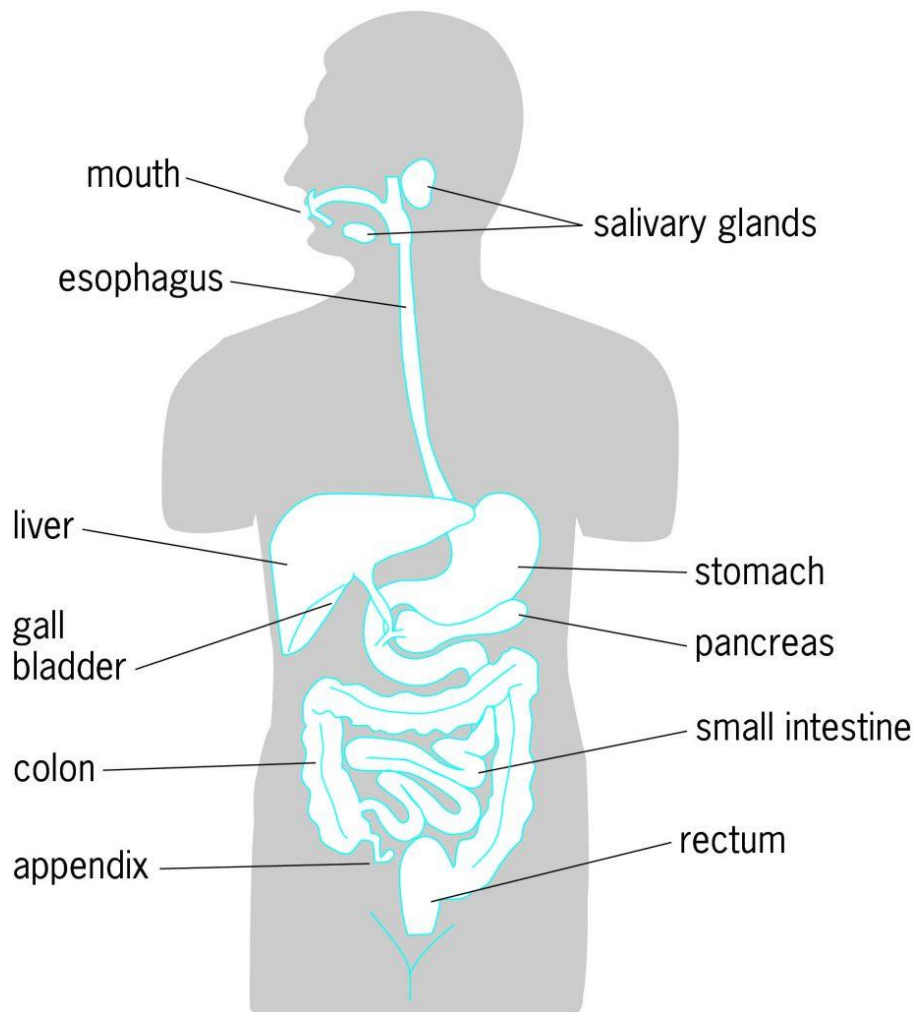
We have been using food additives, such as sugar and salt, for centuries. In recent years, the need for these ingredients has greatly increased due to the **advancement of food science, safety and standards**, the need for foods to be shipped long distances, and a desire for foods to be more shelf stable and affordable.



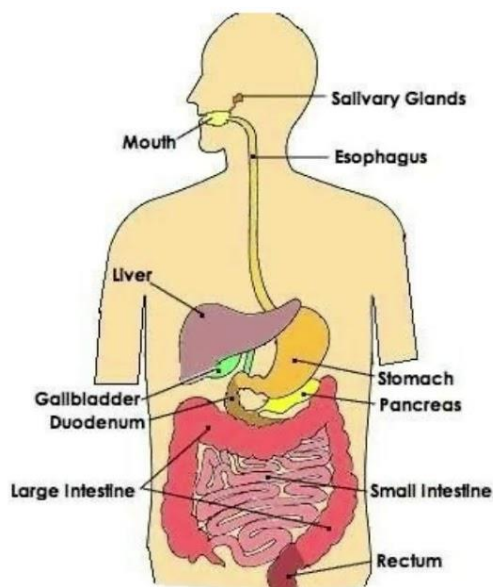
WHY DO FOOD MANUFACTURERS ADD SO MANY OF THESE ARTIFICIAL INGREDIENTS TO THEIR FOODS?

- To improve shelf life
- To improve flavor
- Keep their color
- As a way to make a profit - may own ingredients

STRUCTURE OF DIGESTIVE SYSTEM



PARTS OF THE DIGESTIVE SYSTEM



Digestion starts in the **mouth**. Saliva wets and softens the food and our teeth chew it well. Once it is well chewed, it becomes easier to swallow the food. The food passes through a tube called the **food pipe(esophagus)** into the

Digestive System

1. The Mouth or Buccal Cavity

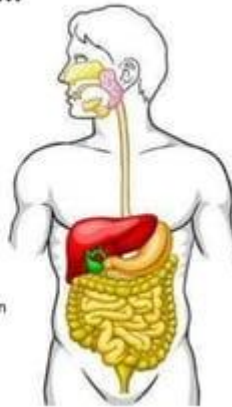
1. Set Of 32 Teeth,
 2. A Tongue
 3. Salivary Glands
- All these parts help in the **chewing** and breakdown of the food particles

2. The Esophagus

- **transportation of food** and liquid from the mouth to the stomach

3. The Stomach and the Pancreas

- **Gastric juices and enzymes** are released in the stomach which aid in the further digestion of the food
- The pancreas secrete **LIPASE & AMYLASE**; they digest carbohydrate & Fats respectively



STRUCTURE OF DIGESTIVE SYSTEM WORKSHEET

HUMAN DIGESTION SYSTEM

Name: _____ Date: _____

Name the parts of the digestion system.

